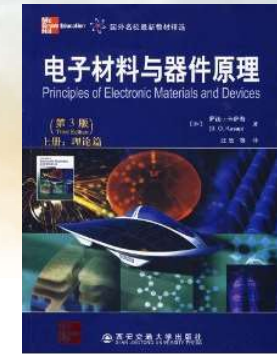
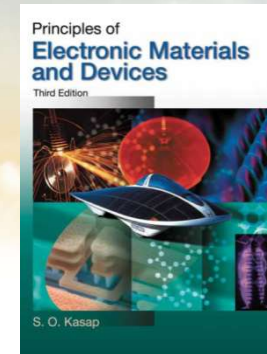




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch3-5 Elementary Quantum Physics

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Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

Four Quantum Numbers

Table 3.3 The four quantum numbers for the hydrogenic atom

n	Principal quantum number	$n = 1, 2, 3, \dots$	Quantizes the electron energy
ℓ	Orbital angular momentum quantum number	$\ell = 0, 1, 2, \dots (n - 1)$	Quantizes the magnitude of orbital angular momentum L
m_ℓ	Magnetic quantum number	$m_\ell = 0, \pm 1, \pm 2, \dots, \pm \ell$	Quantizes the orbital angular momentum component along a magnetic field B_z
m_s	Spin magnetic quantum number	$m_s = \pm \frac{1}{2}$	Quantizes the spin angular momentum component along a magnetic field B_z

Maximum number of electrons for a fixed l : $2(2l+1)$

Maximum number of electrons for a fixed n : $\sum_{i=1}^n 2[2(n-i)+1]$

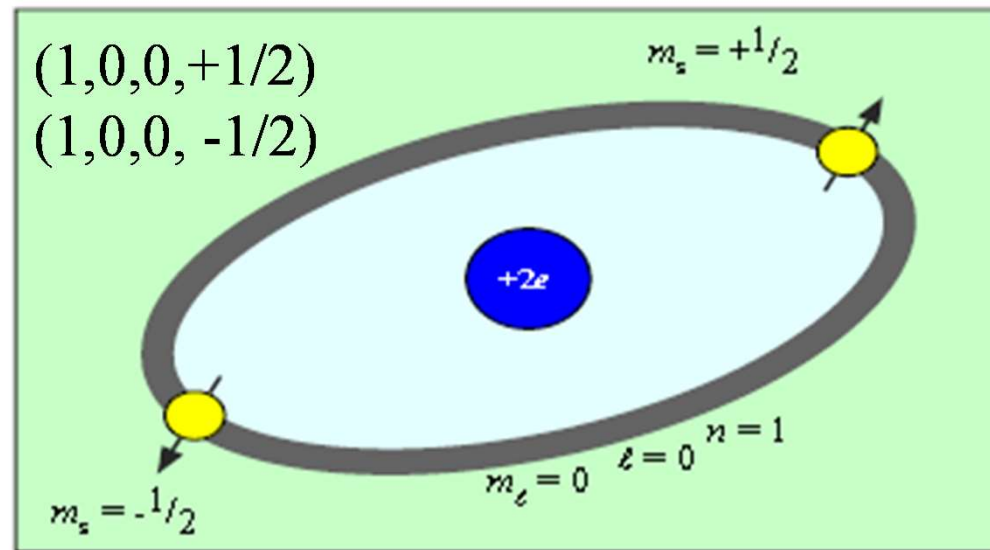
Maximum total number of electrons for a atom with n orbitals: $\sum_{j=1}^n \sum_{i=1}^j 2[2(j-i)+1]$



1. Pauli Exclusion Principle

No more than 2 electrons may occupy any given orbital. If 2 electrons occupy one orbital, then their spin must be paired.

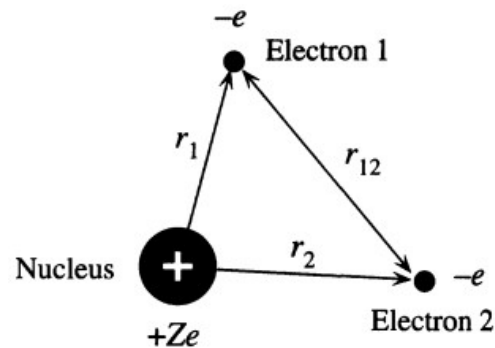
In other word, no two electrons in the same atoms can have exactly the same values for quantum number n , l , m_l , and m_s .



泡利不相容原理：在 n ， l 和 m_l 确定的情况下，每个轨道最多只能容纳两个电子，且自旋相反配对



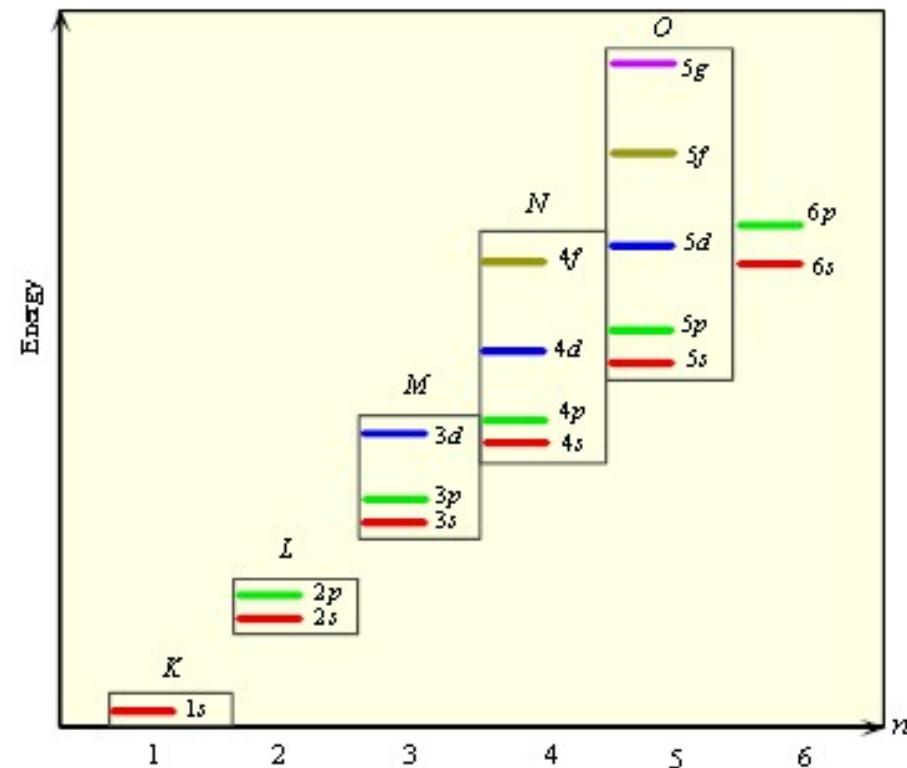
2. Minimum Energy principle



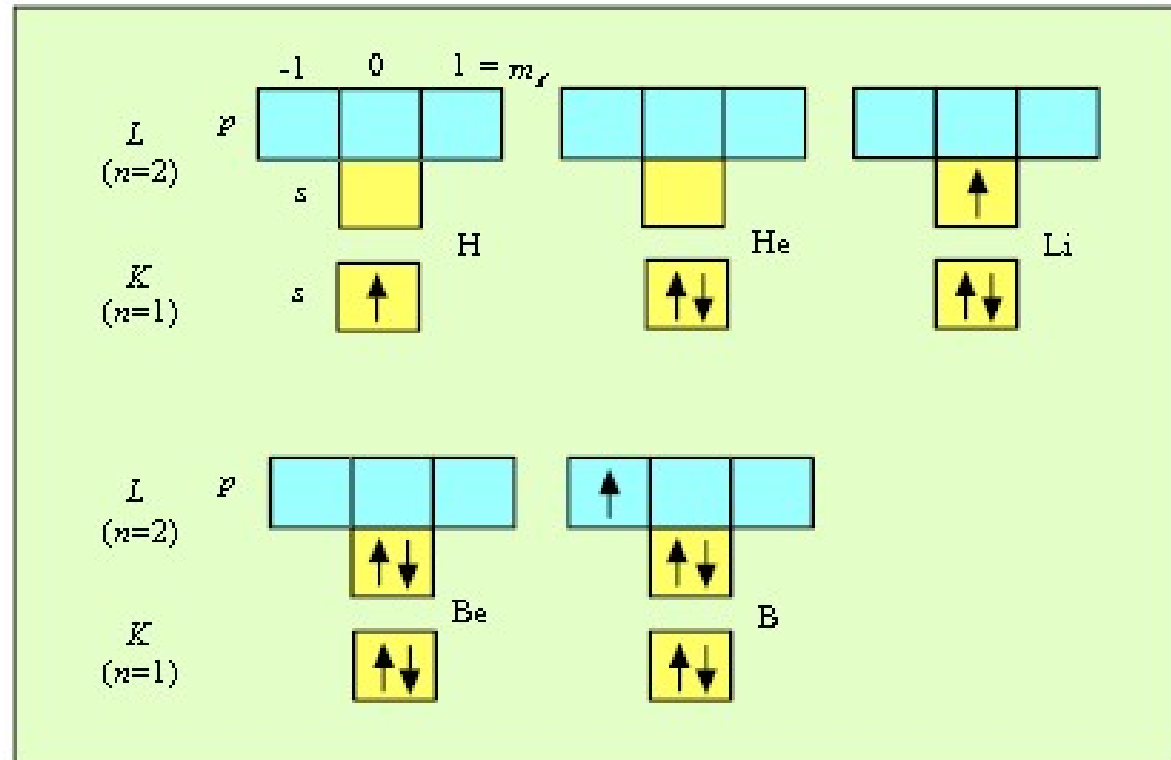
$$V(r_1, r_{12}) = -\frac{2e^2}{4\pi\epsilon_0 r_1} + \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

Due to the second term in the electron-electron potential energy term, the energy of the electron is not only related to n but also related to l .

$$E_{n,l}$$



2. Minimum Energy principle

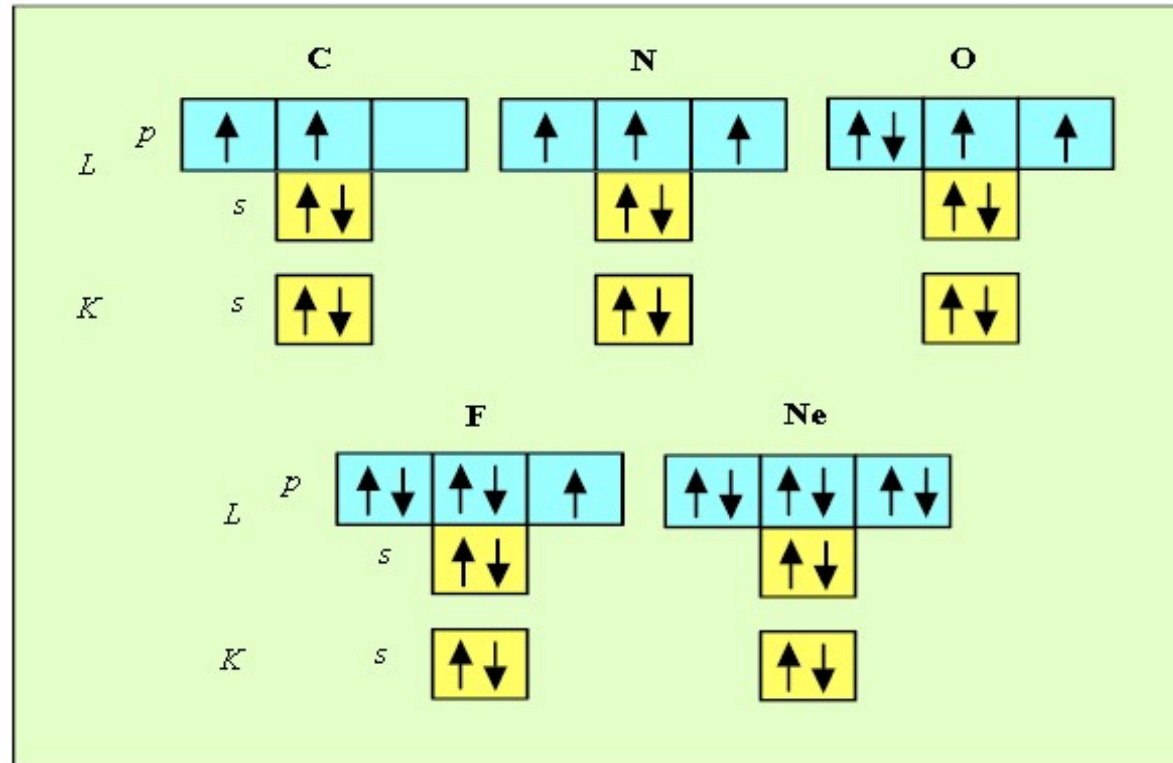


Electronic configurations for the first five elements. Each box represents an orbital



3. Hund's Rule

An atom in its ground state adopts a configuration with the greatest number of unpaired electrons



Electronic configurations for C, N, O, F and Ne atoms. Notice that Hund's rule forces electrons to align their spins in C, N and O. The Ne atom has all the K and L orbitals full.

简并轨道（能级相同的轨道）只有被电子逐一自旋平行地占据后，才能容纳第二个电子



- The lattice is like an apartment, the electronics are like tenants

晶格就像一栋公寓，电子就像住户

- You must pay the additional fee if you want to live in higher level (band gap)

规定从底层入住，跃层选房必须支付费用（能级）

- The layout of the house is different, but each house can accommodate up to two people. It must be a couple (the Pauli exclusion principle)

房屋的格局不同，但每间房子最多可住两人，必须是夫妻（泡利原理）

- When the many houses, you can live in a single room (conduction band)

在住房条件比较宽松时，可以住单间（导带）

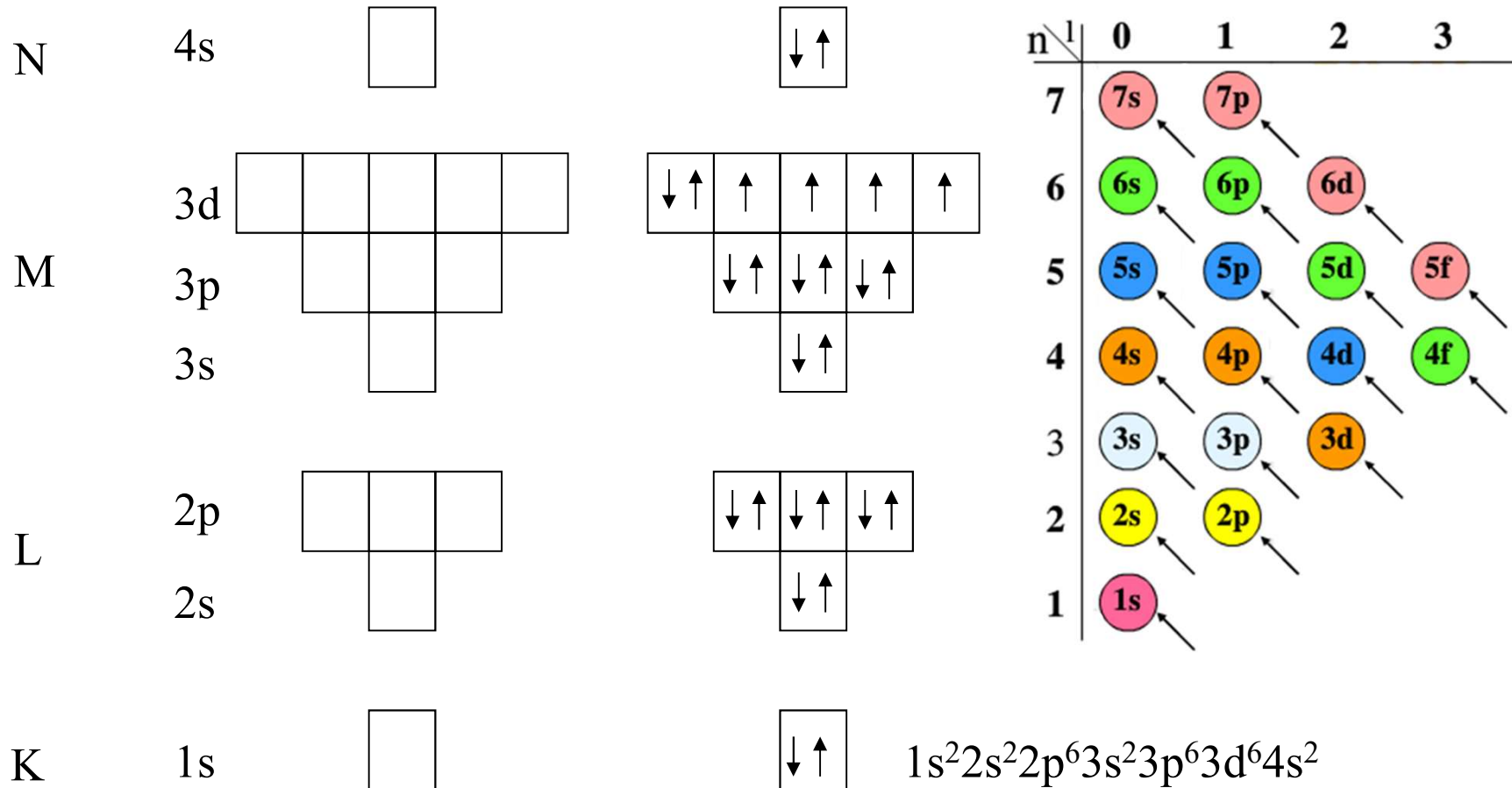
- The residents on the same floor can freely change rooms at any time (band gap)

同层住户随时可自由换房（能带）



Example

Write down the electron configuration of Iron (Fe) which has 26 electrons.



Example

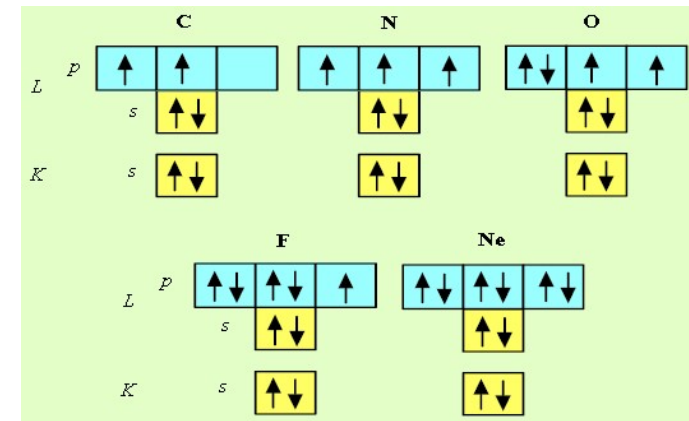
Moreover, the equivalent orbit is more stable **for full, semi-full or full-empty state**, that is, the following electronic structures are more stable:

另外：等价轨道在**全充满**、**半充满**或**全空**的状态是比较稳定的，亦即下列电子结构是比较稳定的：

Full (全充满): p^6 , d^{10} , or f^{14}

Semi-Full (半充满): p^3 , d^5 , or f^7

Full-Empty (全空): p^0 , d^0 , or f^0



$_{47}\text{Ag}$

$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^1$

$_{24}\text{Cr}$

$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$



Pauli Exclusion Principle



The Nobel Prize in Physics 1945
Wolfgang Pauli

The Nobel Prize in Physics 1945



Wolfgang Pauli

The Nobel Prize in Physics 1945 was awarded to Wolfgang Pauli *"for the discovery of the Exclusion Principle, also called the Pauli Principle"*.



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Solvay Conference 1927



SOLVAY CONFERENCE 1927

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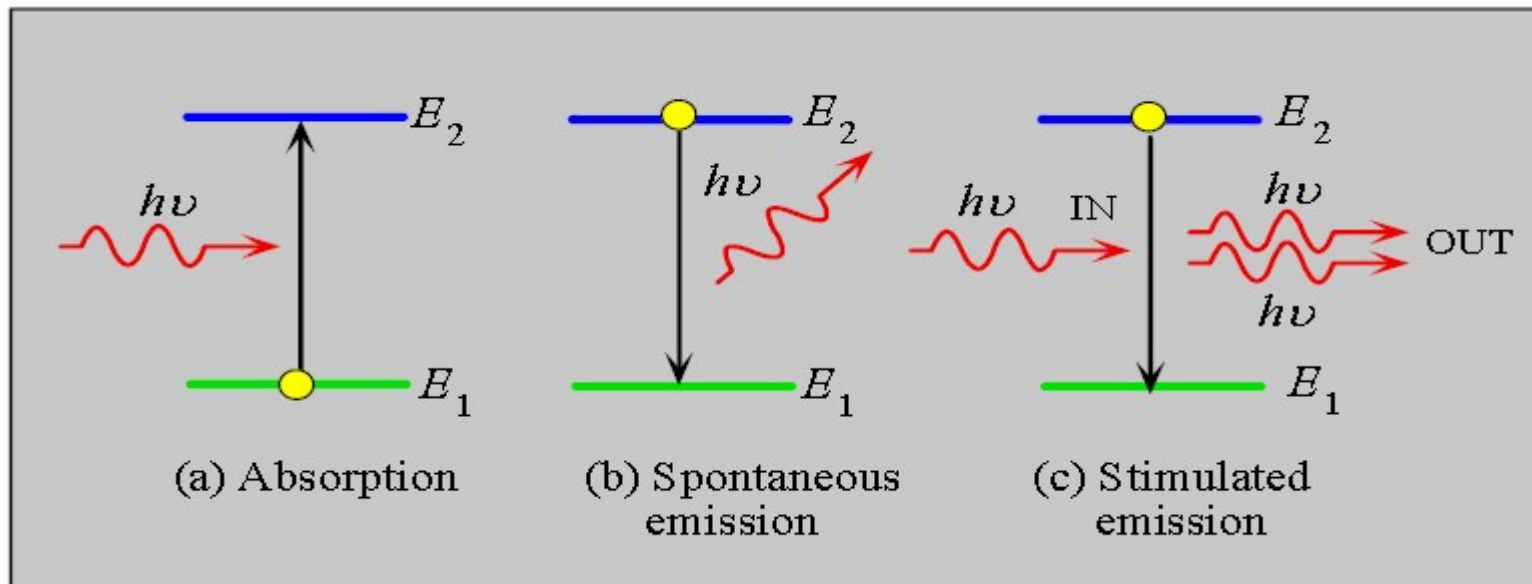


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Laser

Light Amplification by Stimulated Emission of Radiation

Characteristics of laser: Monochromatic; Coherence; Directional; Intensity



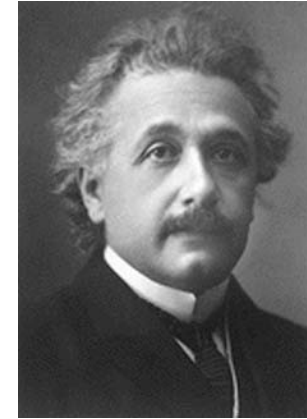
Spontaneous emission: Once an atom is in an excited state, there is a constant probability that it will jump back to the lower state by emitting a photon.

Stimulated emission: An atom is in an excited state and a photon is incident on it. The incoming photon increases the probability that the excited atom will return to the ground.



1917 Stimulated Emission

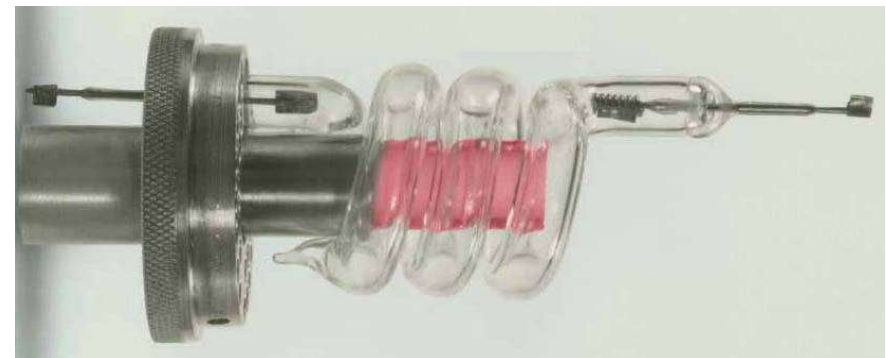
- Einstein proposed the process that makes lasers possible, called stimulated emission. He theorized that, besides absorbing and emitting light spontaneously, electrons could be stimulated to emit light of a particular wavelength
- But it would take nearly 40 years before scientists would be able to amplify those emissions, proving Einstein correct and putting lasers on the path to becoming the powerful and ubiquitous tools they are today.



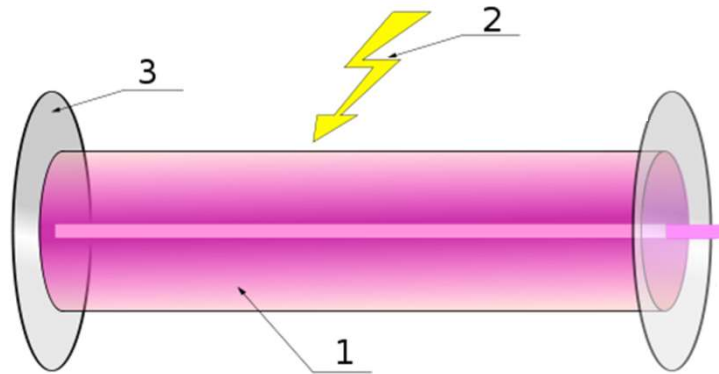
"A splendid light has dawned on me..."
- *Albert Einstein*

1960 First Optical Laser

Theodore H. Maiman, a physicist at Hughes Research Laboratories in Malibu, Calif., constructs the first laser using a cylinder of synthetic ruby measuring 1 cm in diameter and 2 cm long, with the ends silver-coated to make them reflective and able to serve as a Fabry-Perot resonator. Maiman uses photographic flashlamps as the laser's pump source.

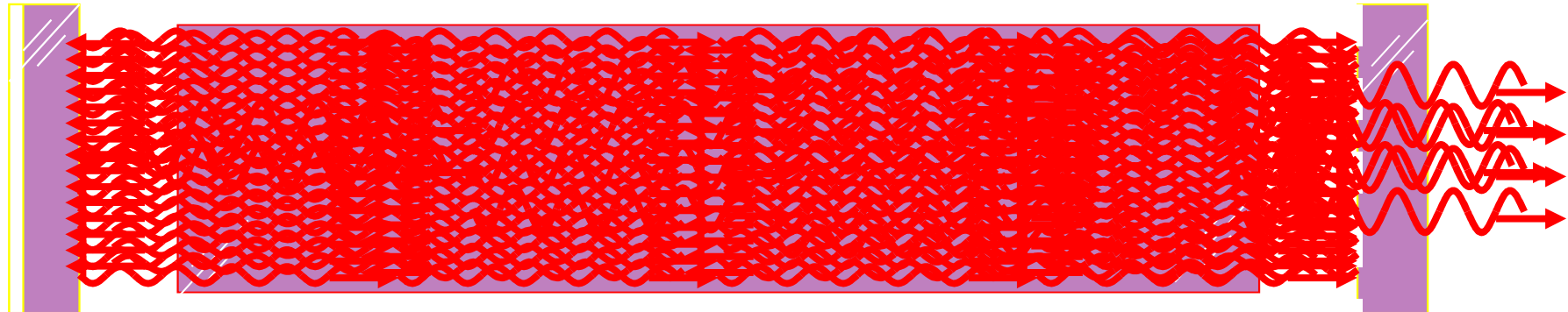


Laser Fundamentals



Principal components

1. Gain medium (增益材料)
2. Pumping (泵浦)
3. Optical cavity (光学谐振腔)



Photons with:

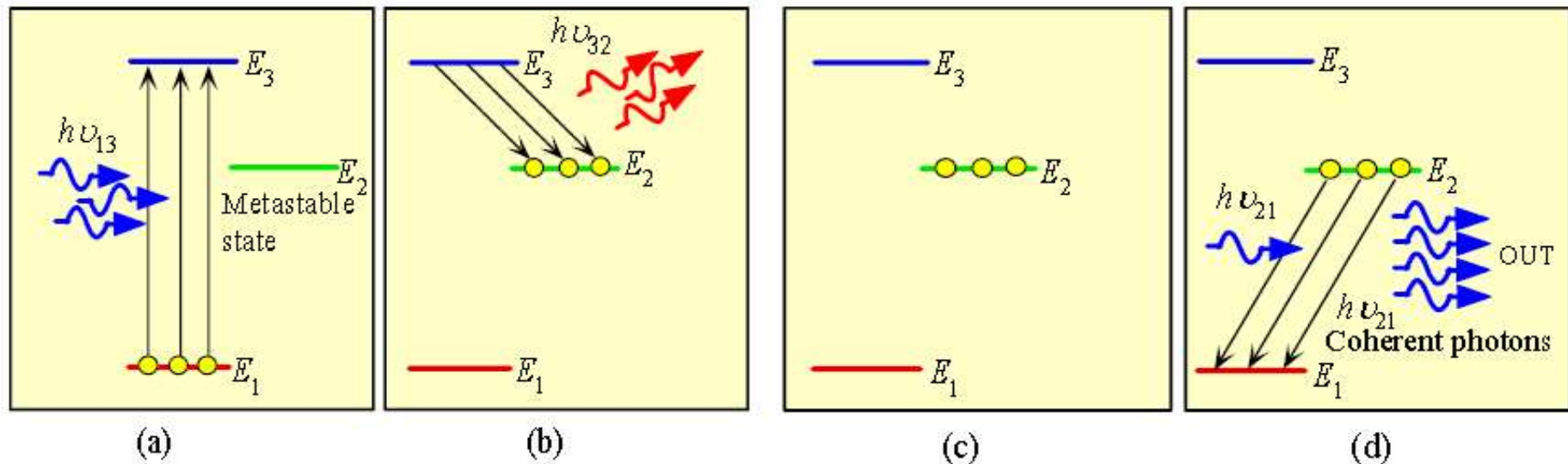
- same energy: **Monochromatic**
- same direction of propagation: **Spatial coherence**
- all in synchrony: **Temporal coherence**



Laser: Population Inversion

When light is incident on a system of atoms, both stimulated absorption and stimulated emission are equally probable.

Generally, a net absorption occurs since most atoms are in the ground state. If you can cause more atoms to be in excited states, a net emission of photons can result.

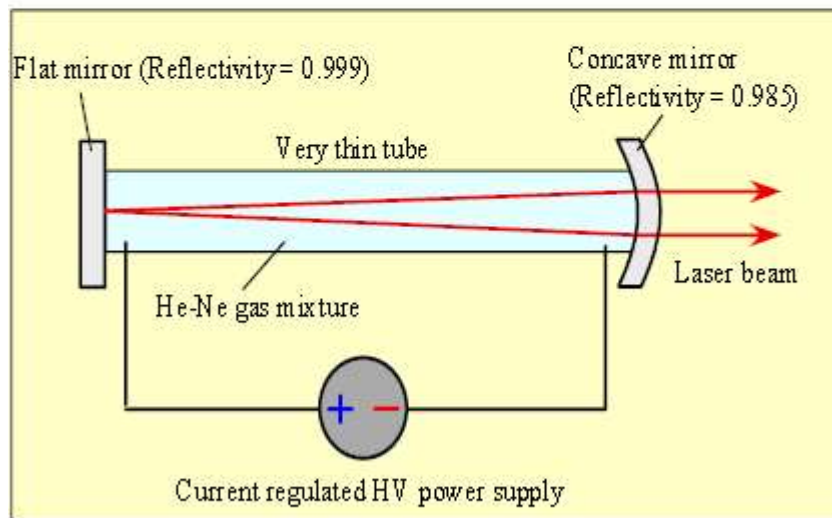
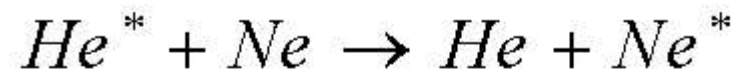
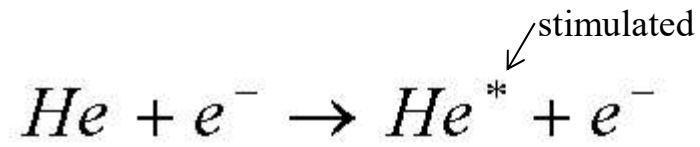


The principle of the LASER. (a) Atoms in the ground state are pumped up to the energy level E_3 by incoming photons of energy $h\nu_{13} = E_3 - E_1$. (b) Atoms at E_3 rapidly decay to the metastable state at energy level E_2 by emitting photons or emitting lattice vibrations. $h\nu_{32} = E_3 - E_2$.

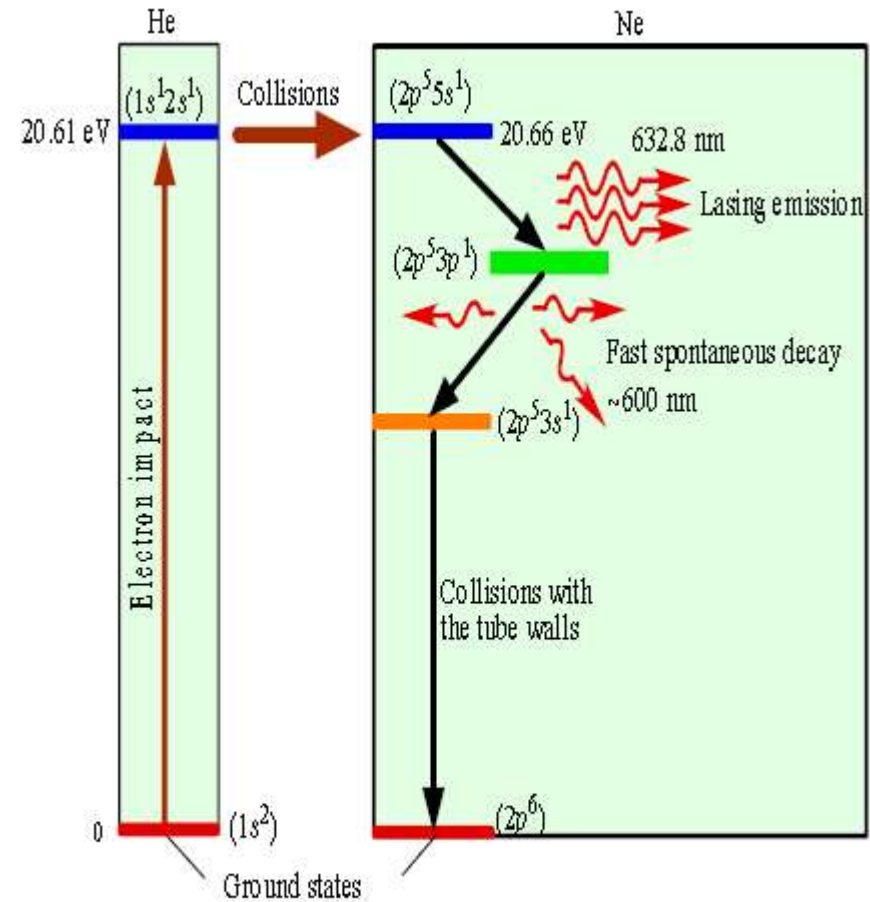
(c) As the states at E_2 are metastable, they quickly become populated and there is a population inversion between E_2 and E_1 . (d) A random photon of energy $h\nu_{21} = E_2 - E_1$ can initiate stimulated emission. Photons from this stimulated emission can themselves further stimulate emissions leading to an avalanche of stimulated emissions and coherent photons being emitted.



He-Ne laser



A schematic illustration of the He-Ne laser



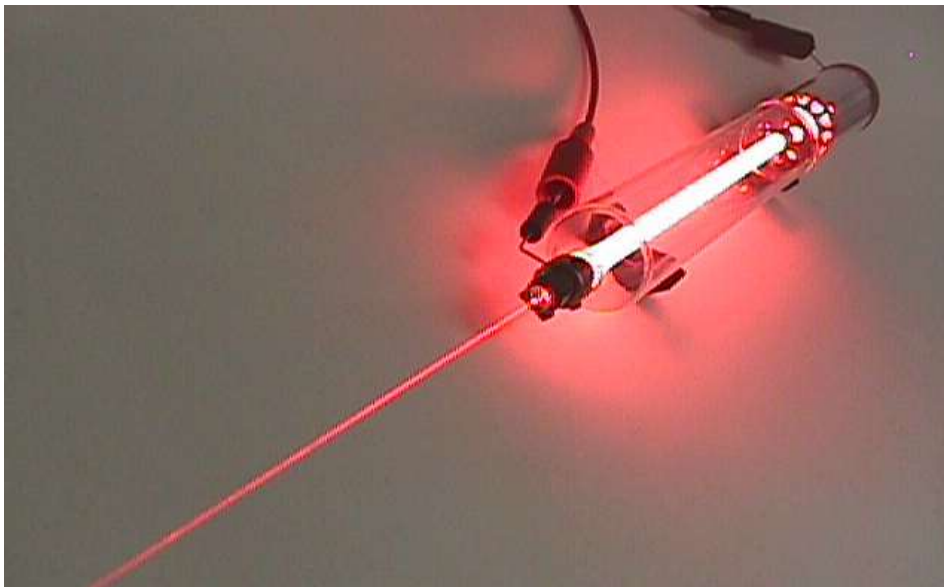
The principle of operation of the He-Ne laser. He-Ne laser energy levels (for 632.8nm emission).

632.8 nm ~~543 nm~~ ~~3.39 μm~~



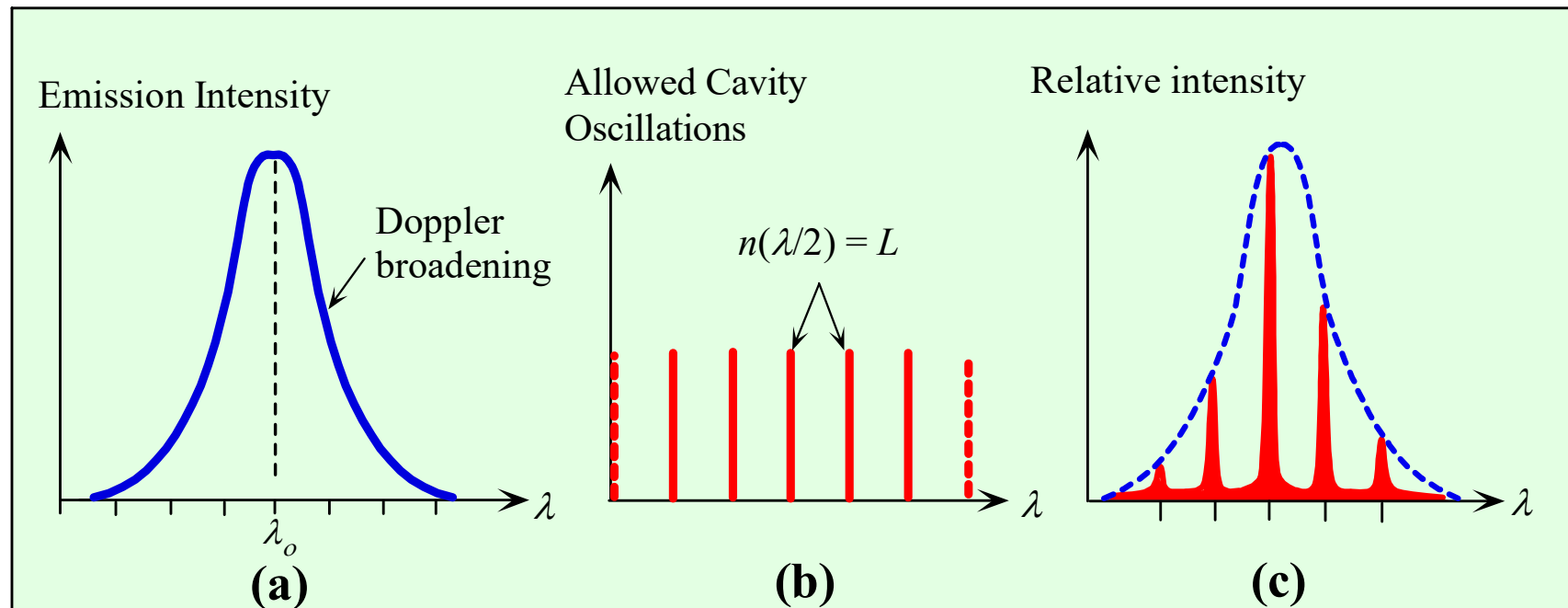
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He-Ne laser



Doppler Broadened Emission

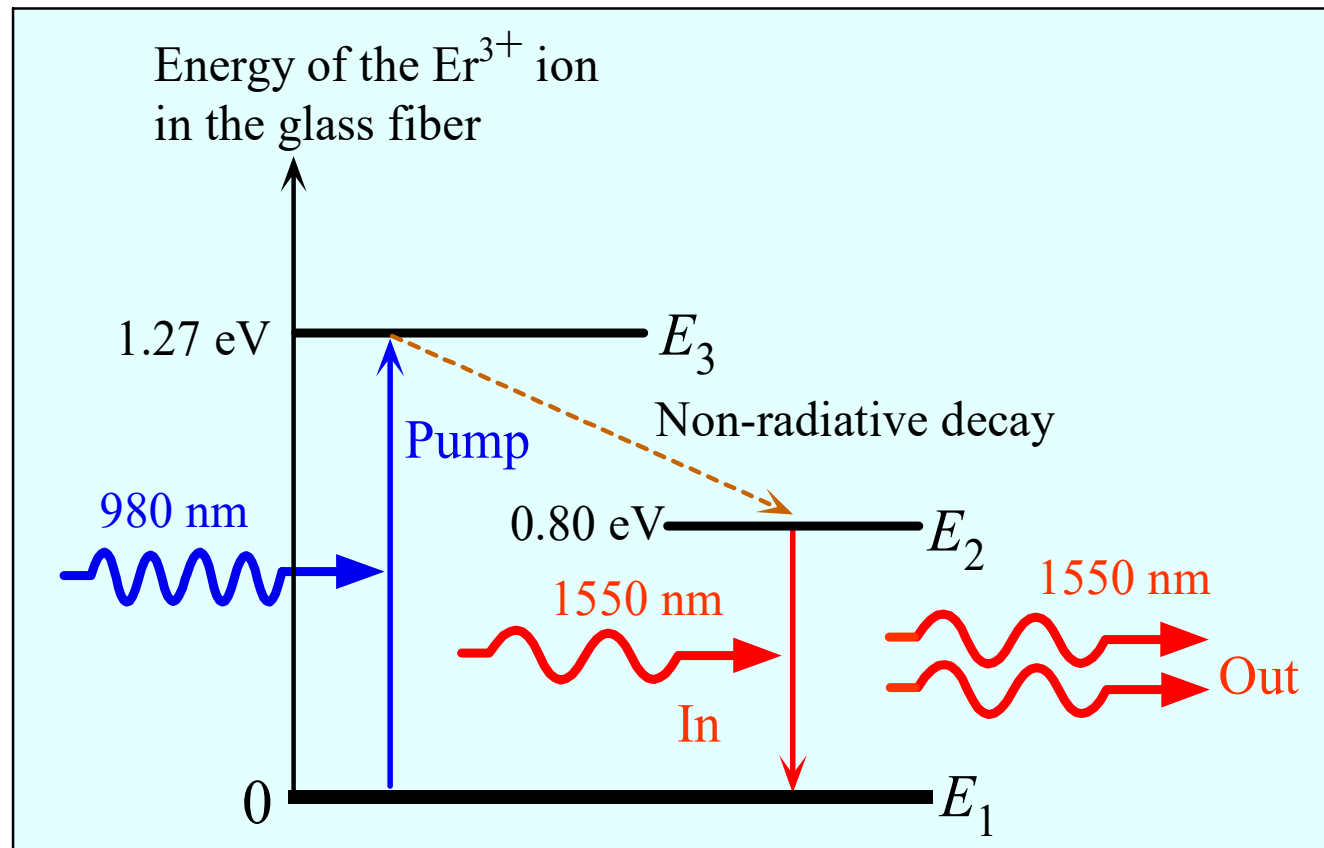
- Move toward an observer, a higher frequency is detected: $\nu_2 = \nu_0 (1 + v_x/c)$
- Move away, a smaller frequency is detected: $\nu_1 = \nu_0 (1 - v_x/c)$
- v_x is the relative velocity of the atom respect to the observer



(a) Doppler broadened emission vs. wavelength characteristics of the lasing medium. (b) Allowed oscillations and their wavelengths within the optical cavity. (c) The output spectrum is determined by satisfying (a) and (b) simultaneously.



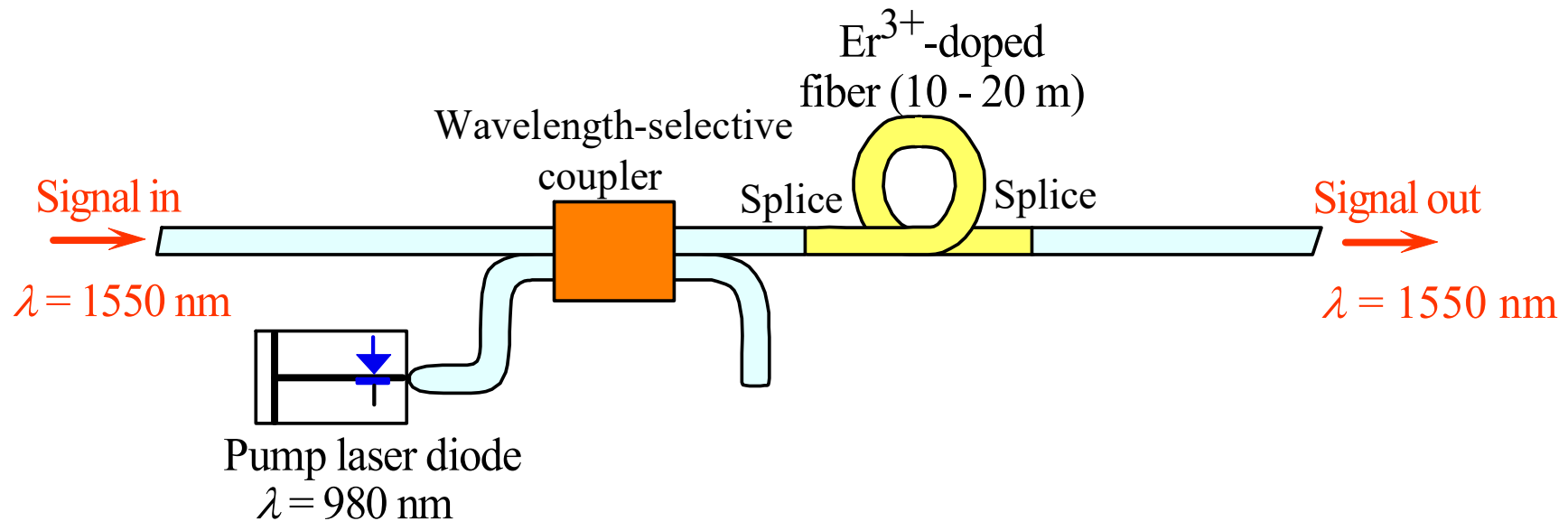
Er³⁺ Doped Fiber Laser (EDFA)



Energy diagram for the Er³⁺ ion in the glass fiber medium and light amplification by stimulated emission from E_2 to E_1 . Dashed arrows indicate radiationless transitions (energy emission by lattice vibrations)



Er³⁺ Doped Fiber Laser (EDFA)

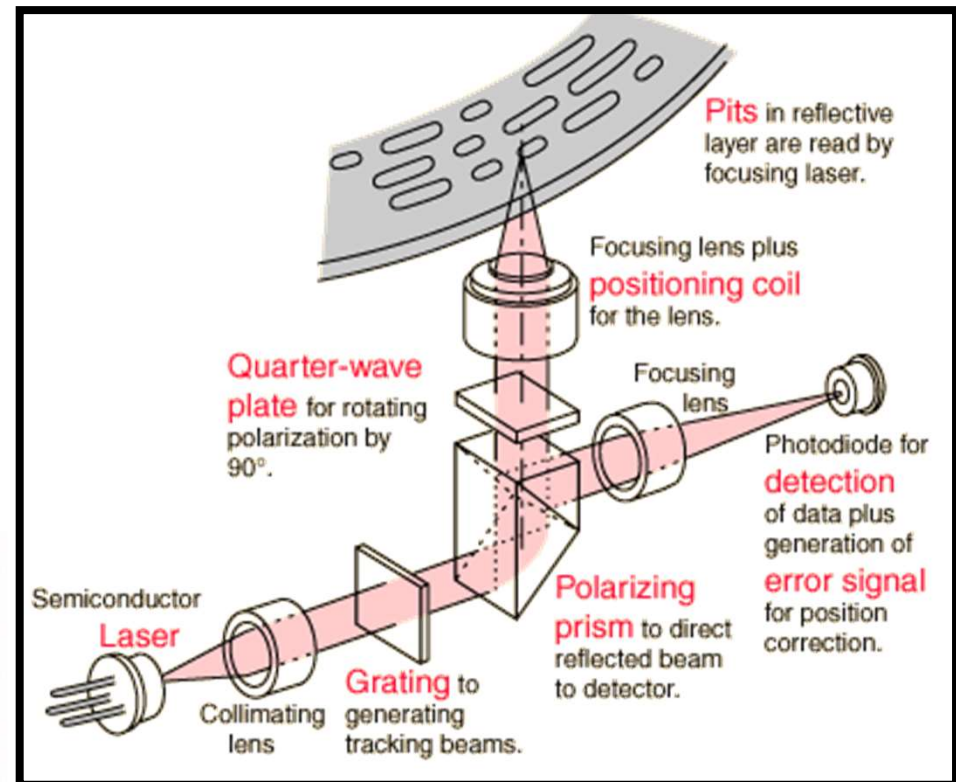
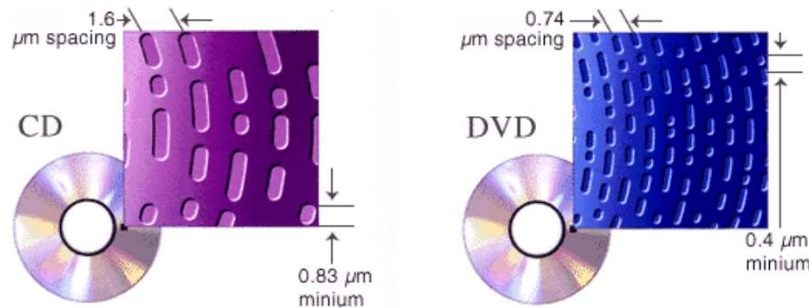
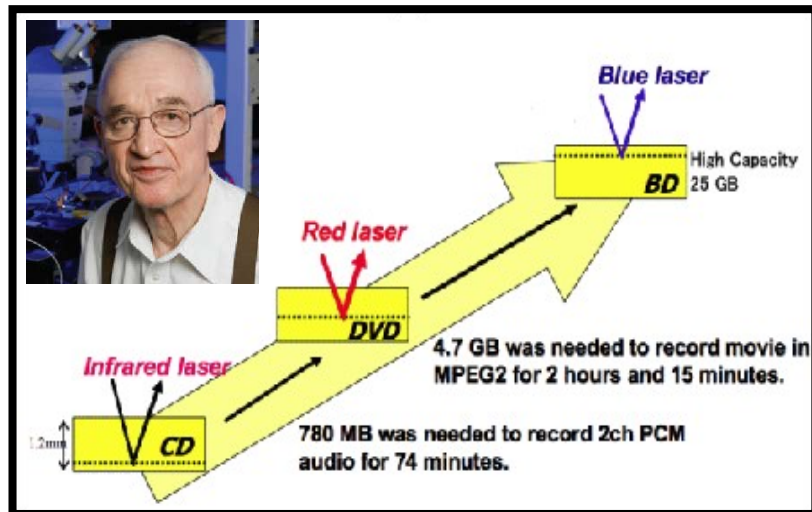


A simplified schematic illustration of an EDFA (optical amplifier). The erbium-ion doped fiber is pumped by feeding the light from a laser pump diode, through a coupler, into the erbium ion doped fiber.



Typical Application of Laser

The detection of the binary data stored in the form of pits on the compact disc is done with the use of a **semiconductor laser**. The laser is focused to a diameter of about 0.8 mm at the bottom of the disc, but is further focused to about 1.7 micrometers as it passes through the clear plastic substrate to strike the reflective layer. The reflected laser will be detected by a photodiode. Moral of the story: without optoelectronics there will no CD player!

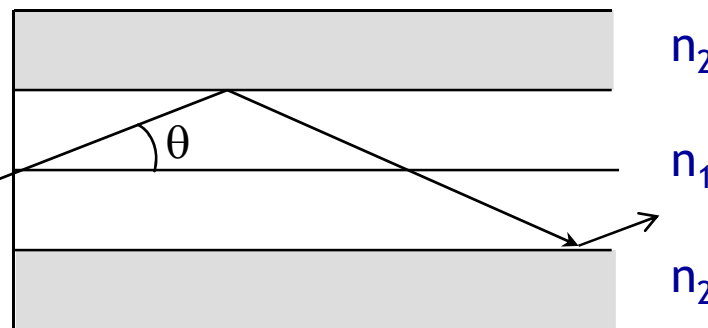
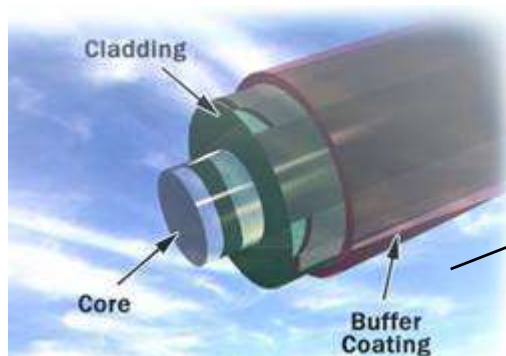


1966 Break Through of Fiber Optics

Charles K. Kao, working with George Hockham at Standard Telecommunication Laboratories in Harlow, England, makes a discovery that leads to a breakthrough in fiber optics. He calculates how to transmit light over long distances via optical glass fibers, deciding that, with a fiber of purest glass, it would be possible to transmit light signals over a distance of 100 km, compared with only 20 m for the fibers available in the 1960s. Kao receives a 2009 Nobel Prize in physics for his work.



04-Nov-1933 to 23-Sep-2018

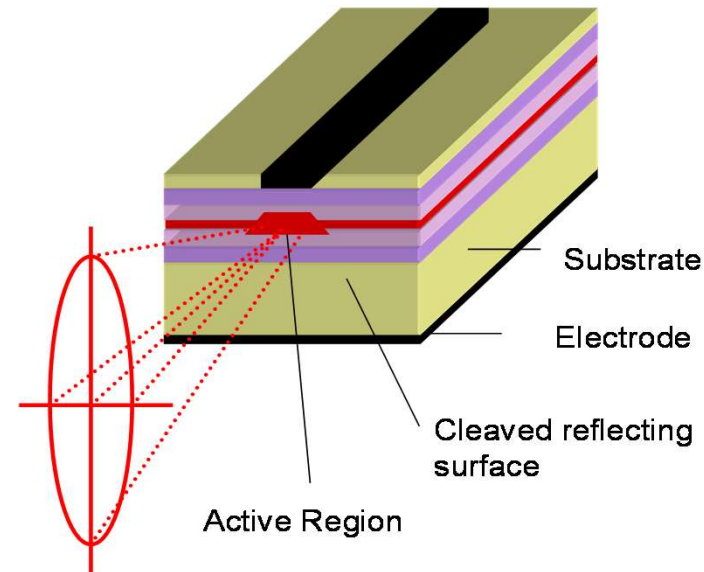
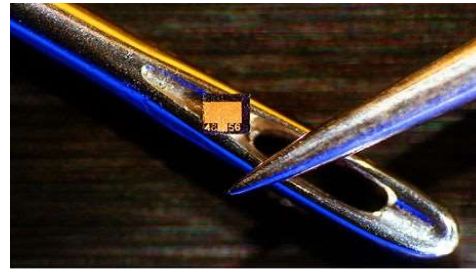


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1970 First Semiconductor Laser

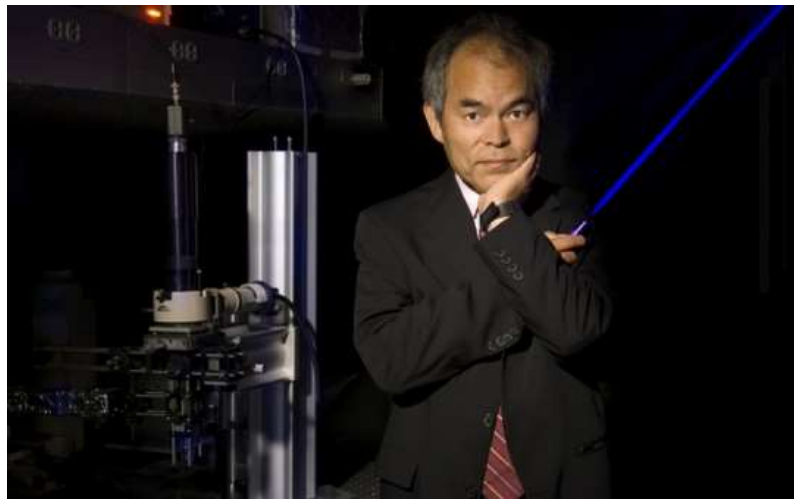
Zhores Alferov's group at the Ioffe Physico-Technical Institute in Russia and Mort Panish and Izuo Hayashi at Bell Labs produce the first CW RT semiconductor lasers, paving the way toward commercialization of fiber optics communications.

2000 Nobel Laureate
in Physics



1997 GaN Blue LEDs

Shuji Nakamura, Steven P. DenBaars and James S. Speck at the University of California, Santa Barbara, announce the development of a gallium-nitride (GaN) laser that emits bright blue-violet light in pulsed operation.



Professor @
University of
California,
Santa Barbara



The 2014 Nobel Prize for Physics has been awarded to Isamu Akasaki, Hiroshi Amano and Shuji Nakamura for their development of blue LEDs.



The Nobel Prize in Physics 2018



Ill. Niklas Elmehed. © Nobel Media

Arthur Ashkin

Prize share: 1/2



Ill. Niklas Elmehed. © Nobel Media

Gérard Mourou

Prize share: 1/4



Ill. Niklas Elmehed. © Nobel Media

Donna Strickland

Prize share: 1/4

The Nobel Prize in Physics 2018 was awarded "for groundbreaking inventions in the field of laser physics" with one half to Arthur Ashkin "for the optical tweezers and their application to biological systems", the other half jointly to Gérard Mourou and Donna Strickland "for their method of generating high-intensity, ultra-short optical pulses."



The Nobel Prize in Physics 2023



III. Niklas Elmehed © Nobel
Prize Outreach
Pierre Agostini
Prize share: 1/3

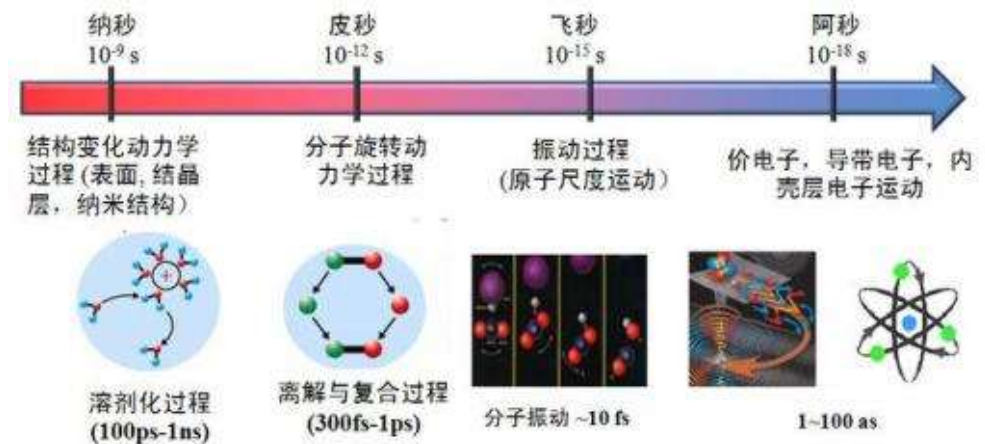


III. Niklas Elmehed © Nobel
Prize Outreach
Ferenc Krausz
Prize share: 1/3



III. Niklas Elmehed © Nobel
Prize Outreach
Anne L'Huillier
Prize share: 1/3

The Royal Swedish Academy of Sciences has decided to award the Nobel Prize in Physics 2023 to Pierre Agostini, Ferenc Krausz, and Anne L'Huillier "for experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter".

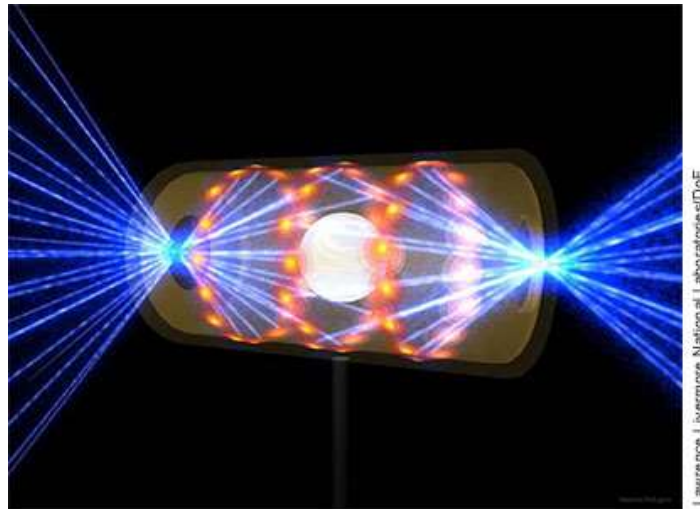


The Use of Lasers

June 2009: NASA launches the Lunar Reconnaissance Orbiter (LRO). LOLA, the Lunar Orbiter Laser Altimeter on the LRO, will use a laser to gather data about the high and low points on the moon. NASA will use that information to **create 3-D maps** that could help determine lunar ice locations and safe landing sites for future spacecraft.



While orbiting the moon, the Lunar Reconnaissance Orbiter will take pictures and gather information about the moon's surface.



Artist's rendition showing a National Ignition Facility target pellet inside a hohlraum capsule with laser beams entering through openings on either end. The beams compress and heat the target to the necessary conditions for nuclear fusion to occur.

January 2010: The National Nuclear Security Administration announces that NIF has successfully delivered a historic level of laser energy — more than 1 MJ — to a target in a few billionths of a second and demonstrated the target drive conditions required to achieve **fusion ignition**, a project scheduled for the summer of 2010. The peak power of the laser light is about 500 times that used by the US at any given time.

